

Sameer Shashikant Rajendra

Jersey City, NJ | (201) 927-4722 | sameer.rajendra@outlook.com | [LinkedIn](#) | [GitHub](#) | STEM OPT Eligible

EDUCATION

Stevens Institute of Technology – Hoboken, NJ <i>MS in Applied Artificial Intelligence</i>	May 2026
Symbiosis Skills and Professional University – Pune, India <i>Bachelor of Technology in Mechatronics</i>	May 2021

TECHNICAL SKILLS

Languages: Python, C++, CUDA C/C++

GPU & Parallel Computing: CUDA, cuDNN, FlashDecoding / split-KV, NVML, NVTX, PyBind11, Nsight Compute (ncu), warp-level primitives, shared-memory tiling, roofline / bandwidth analysis

Deep Learning Frameworks: PyTorch (FSDP), FlashAttention-2, TensorFlow, HuggingFace Transformers, TorchVision, NSGA-II

Inference & Model Optimization: Grouped-Query Attention (GQA), split-KV / FlashDecoding, online softmax, kernel fusion, sparse / top-k attention (ablation), KV-cache quantization (in progress), LoRA, vLLM

Computer Vision: EfficientNet, OpenCV, CuPy GPU preprocessing,

Tools & Platforms: Docker, Linux, Git, QuestDB, Jupyter, SLURM

WORK EXPERIENCE

Oneirix Labs Pvt. Ltd. – Pune, India Associate Engineer	Sept 2021 – June 2024
- Built a multi-task computer vision system on recorded Pune Metro CCTV footage four concurrent inference tasks (crowd density, pose-based intrusion/track-jump detection, property damage, unattended baggage), using PyTorch and TorchVision. - Owned the controller↔scheduler communication subsystem for a smart traffic-light deployment: wrote the bidirectional C++ protocol decode/forward scripts and packaged the full stack into a Docker container for on-site deployment - Built an experimental telemetry rig for motor-aging characterization instrumented Hall, load, and temperature sensors to record speed-torque curves across load conditions, then correlated curve deviations against thermal degradation of coils and magnets. - Contributed a coronary-artery graph traversal pipeline for stenosis screening CNN-based main-vessel identification, BFS traversal for secondary/tertiary branches, and geometric diameter-reduction detection on the resulting graph.	

PROJECTS

LLM Inference Optimization GitHub Report	Mar 2026 – Present
- Built a split-KV (FlashDecoding-style) fused GQA decode kernel in C++/CUDA (FP16, Hopper sm_90a) for Llama-3.1-8B — partitions the KV sequence across all SMs with an exact log-sum-exp combine; numerically exact vs PyTorch SDPA across all 32 layers (max abs error $\leq 1e-3$ up to 64K context) and 18× faster than a naïve single-block decode kernel, exposed to PyTorch via PyBind11. - Benchmarked the kernel against FlashAttention with Nsight Systems, measuring ~6% vs ~77% of H100 HBM bandwidth and diagnosing the gap to asynchronous memory pipelining; the kernel uses dynamic shared-memory allocation (bypassing the 48 KB static limit) and half2 vectorization, and is numerically exact across all 32 layers (max abs error $\leq 1e-3$). - Profiled end-to-end decode to show attention is only ~14% of batch-1 latency (weight-GEMMs and launch overhead dominate), redirecting optimization toward KV-cache quantization and query-aware sparsity as the byte-reducing levers for long-context decode. - Wrote a PyTorch FSDP + custom LoRA (rank=16, $\alpha=32$) single-node multi-GPU fine-tuning harness with NCCL backend and auto-wrap policy; cross-validated kernel numerics against a JAX / Pallas reference implementation for framework-independent verification.	

Adaptive Evolutionary GPU Optimization Framework (AEGOF) | [GitHub](#) | [Report](#)

Sept – Dec 2025

- Architected an **NSGA-II multi-objective evolutionary autotuner** jointly optimizing 8 hardware / software knobs — power cap, batch size, sequence length, **bf16 vs. fp16 precision**, kernel-fusion mode, block size, tile configuration, and **FlashAttention-2 vs. CUTLASS attention backends** — for Llama-2-7B inference on **NVIDIA H100**.
- Evolved a Pareto front of **throughput vs. power** configurations across 16 individuals × 5 generations, evaluating 238+ unique configurations and reaching an **8.09× efficiency multiplier** (tokens / sec / Watt) over PyTorch defaults; results serialized to a Hall-of-Fame JSON for reproducible deployment.
- Instrumented a white-box benchmark harness measuring **tokens/sec throughput, NVML power draw, and Model FLOPs Utilization (MFU)** per trial, with robust CUDA OOM and subprocess-timeout handlers enabling unattended overnight evolutionary runs without manual intervention.

Diabetic Retinopathy Severity Grading (EfficientNet-B4) | [GitHub](#) | [Report](#)

Jan 2025 – May 2025

- Built an **EfficientNet-B4 transfer-learning pipeline** (ImageNet-pretrained backbone, fine-tuned classifier head with dropout 0.3) for 5-class diabetic retinopathy severity grading on 380×380 retinal fundus images, reaching **80.49% validation accuracy** and **0.91 F1 on the dominant class**.
- Engineered a **CuPy GPU-accelerated data loader** for high-throughput preprocessing of retinal images, and integrated sklearn-balanced class weights into nn.CrossEntropyLoss to address severe class imbalance across grades 0–4 (most patients are No-DR).

Financial News Sentiment Classification Pipeline | [GitHub](#) | [Report](#)

Jan – May 2025

- Built an end-to-end pipeline from QuestDB ingestion through GPT-Neo 125M fine-tuning to on-demand inference, processing 36,000+ financial news articles with 92% accuracy and 0.72 macro F1-score.
- Engineered a custom WeightedTrainer with class-imbalanced loss to handle severe label skew; packaged trained model and tokenizer into a streamlined inference function for real-time sentiment scoring.

AWARDS & RECOGNITION

HACK MIT-WPU 2024, CCTV Video Analytics Hackathon

May 2024

Maha Metro × Ministry of Education, Maharashtra. Built a real-time multi-task CCTV analytics system for Pune Metro recognized for innovation in sustainable urban transit infrastructure.